

Quiz 6

NAME: _____

Q.1. $f(x) = 2x^2 + bx + c$ is some quadratic equation with unknown coefficients b, c . Its vertex is $(-1, -1)$. Find the coefficients b and c . (**Hint:** start by setting up the vertex form equation)

Q.2. Find all x -intercepts of the circle with center $(4, 2)$ and radius 2. (**Hint:** start by setting up the equation of the circle)

Q.3. Suppose we have a pair of functions $f(x) = x^2 - 2x + 1$ and $g(x) = x - 1$. First, state the domain of the function $\frac{f(x)}{g(x)}$ and then simplify the rational expression $\frac{f(x)}{g(x)}$.

Q.4. Suppose we have a pair of functions $f(x) = \sqrt{x}$ and $g(x) = 2x + 4$. Compute the composition $(f \circ g)(x)$ and state the domain of this composite function in interval notation.